

EECE 306 Lab 2 notebook.

Coordinate Systems, Points and Vector Components

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Fill in every TODO, run `publish('Lab02_notebook.m')` from the toolbox root, print the HTML to PDF, three to five pages.

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Setup

```
clear; close all;  
rng(306);
```

Point roundtrips, 1000 random points

```
r = randn(1000,3);  
esph = max(em.vec.mag(em.coord.sph2c(em.coord.c2sph(r)) - r));  
ecyl = max(em.vec.mag(em.coord.cyl2c(em.coord.c2cyl(r)) - r));  
fprintf('spherical roundtrip max error = %.3e\n', esph);  
fprintf('cylindrical roundtrip max error = %.3e\n', ecyl);
```

```
spherical roundtrip max error = 5.947e-15  
cylindrical roundtrip max error = 2.047e-15
```

Poles and the phi convention

```

disp('c2sph([0 0 1]) ='); disp(em.coord.c2sph([0 0 1]))
disp('c2sph([0 0 -1]) ='); disp(em.coord.c2sph([0 0 -1]))
q = [ 1 1 0; -1 1 0; -1 -1 0; 1 -1 0 ];
p = em.coord.c2cyl(q);
fprintf('phi in the four quadrants = %.4f %.4f %.4f %.4f rad\n', p(:,2));
fprintf('all inside [0, 2*pi) = %d\n', all(p(:,2) ≥ 0 & p(:,2) < 2*pi));

```

```

c2sph([0 0 1]) =
    1    0    0
c2sph([0 0 -1]) =
    1.0000    3.1416    0
phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
all inside [0, 2*pi) = 1

```

Vector component roundtrips and the invariant

```

A = randn(1000,3);
As = em.coord.vecC2Sph(A, r);
ev = max(em.vec.mag(em.coord.vecSph2C(As, r) - A));
em_ = max(abs(em.vec.mag(As) - em.vec.mag(A)));
fprintf('vector roundtrip (spherical pair) max error = %.3e\n', ev);
fprintf('magnitude change under conversion max = %.3e\n', em_);
Ac = em.coord.vecC2Cyl(A, r);
ev2 = max(em.vec.mag(em.coord.vecCyl2C(Ac, r) - A));
fprintf('vector roundtrip (cylindrical pair) max error = %.3e\n', ev2);

```

```

vector roundtrip (spherical pair) max error = 8.882e-16
magnitude change under conversion max = 8.882e-16
vector roundtrip (cylindrical pair) max error = 6.280e-16

```

The two position demonstration

The same Cartesian vector $A = 2x + 3y$, expressed in spherical components at two different positions.

```

A0 = [2 3 0];
disp('at (0, 5, 0), (Ar, Atheta, Aphi) ='); disp(em.coord.vecC2Sph(A0, [0 5 0]))
disp('at (4, 0, 0), (Ar, Atheta, Aphi) ='); disp(em.coord.vecC2Sph(A0, [4 0 0]))
% TODO two to four sentences. The vector did not change. Explain exactly
% what did change, and why a position argument is therefore mandatory for
% component conversion and meaningless for point conversion.
%
% The vector did not change, but the spherical components change.
% This is because Cartesian components don't depend on the current
% position, but spherical components do.

```

```

at (0, 5, 0), (Ar, Atheta, Aphi) =
    3.0000e+00    1.8370e-16   -2.0000e+00
at (4, 0, 0), (Ar, Atheta, Aphi) =
    2.0000e+00    1.2246e-16    3.0000e+00

```

Interpretation

TODO what class of bug does the roundtrip test catch, and what class does it miss. Name the invariant that catches what the roundtrip misses, and say why it is reference free.

the roundtrip test catches errors when converting from Cartesian $\rightarrow$ spherical $\rightarrow$ Cartesian. This however does not catch if the functions are implemented correctly. This means it will test that our conversions are accurate but not if we made a fatal mistake.

Problems encountered

TODO honest account, or NONE.

Similar to Lab01, getting setup was really the main issue here. It took a while before I was at the point where I was ready to work on the labs (and understand what I'm doing,) but once I understood the task better, completing it was quick.

Big thanks to Carissa and Amelia for helping me get caught up for general information around the file structure and navigating the resources!

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
-----
11.3.0
eps0 = 8.8541878128e-12 F/m
mu0  = 1.2566370614e-06 H/m
c0    = 2.9979245808e+08 m/s
eta0  = 376.730314 ohm
Derived constant checks passed.
mag([3 4 0]) = 5 (expect 5 exactly)
unit([3 4 0]) =
    0.6000    0.8000         0
unit([0 0 0]) (expect all NaN) =
    NaN    NaN    NaN
angle perpendicular = 1.5707963267949 (expect pi/2)
angle antiparallel = 3.14159265358979, real = 1 (expect pi, 1)
fromTo([1 1 1],[2 3 4]) =
    1    2    3
mag    input 100x3 output [100 1]
unit   input 100x3 output [100 3]
```

```
angle input 100x3 output [100 1]
fromTo input 100x3 output [100 3]
PASS test_lab01 0.01 s
spherical roundtrip max error = 5.947e-15
cylindrical roundtrip max error = 2.047e-15
c2sph([0 0 1]) =
    1    0    0
c2sph([0 0 -1]) =
    1.0000    3.1416    0
phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
all inside [0, 2*pi) = 1
vector roundtrip (spherical pair) max error = 8.882e-16
magnitude change under conversion max = 8.882e-16
vector roundtrip (cylindrical pair) max error = 6.280e-16
at (0, 5, 0), (Ar, Atheta, Aphi) =
    3.0000e+00    1.8370e-16   -2.0000e+00
at (4, 0, 0), (Ar, Atheta, Aphi) =
    2.0000e+00    1.2246e-16    3.0000e+00
PASS test_lab02 0.01 s
-----
2 passed, 0 failed, 2 total
All tests passing.
```

[Published with GNU Octave 11.3.0](#)